

SYSTEM PROPOSAL

GreenCare - Virtual Plant Care Journal & Assistant

Mobile Application

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September 8, 2026

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1. Introduction

Background of the Project

Plant care has become increasingly popular among households, students, and urban dwellers as more people recognize the benefits of having plants at home. However, many plant enthusiasts struggle to maintain their plants due to busy schedules, forgetfulness, or lack of knowledge about proper care requirements. Beginners often find it difficult to determine the correct watering frequency, light requirements, and fertilizing schedules for different plant species.

Current Process or System in Use

Currently, most plant owners rely on memory, manual notes, or generic guidelines found online. Some use basic calendar reminders, but these do not adapt to the specific needs of each plant species or account for environmental factors such as seasonality and weather changes. There is no centralized system to track multiple plants, log care activities, and receive timely reminders in one place.

Problems or Gaps Identified

- Plant owners often forget when to water, fertilize, or repot their plants, leading to poor plant health or death.
- Care guidelines found online are generic and not personalized to the specific plants a user owns.
- No organized way to track plant growth history, making it difficult to identify patterns of health decline or improvement.
- Beginners lack easy access to accurate species-specific care information.

Proposed Solution

To address these gaps, the proponents propose GreenCare, a web-based application that allows users to manage their personal plant collection, receive automated care reminders, log maintenance activities, and access species-specific plant information through integrated plant database APIs.

2. Purpose of the Project

This proposal aims to present a web-based application that will address plant care scheduling and plant information management problems, streamline plant maintenance tracking, and improve plant health monitoring and user knowledge for plant enthusiasts, beginner gardeners, and home gardeners.

3. Objectives

3.1 General Objective

To design and develop a web-based plant care management system that helps plant owners efficiently track, schedule, and maintain the care requirements of their personal plant collection.

3.2 Specific Objectives

- To design and develop a web-based system that enables users to manage their plant collection, receive care reminders, and access plant species information.
- To provide Registered Users with the ability to add and manage their plants, log care activities, and view personalized care schedules.
- To provide System Administrators with the ability to manage plant database content, moderate user reports, and configure system settings.
- To automate care scheduling and reminders and reduce instances of missed watering, fertilizing, and other maintenance tasks.
- To ensure the system is secure, scalable, responsive across devices, and easy to maintain and update.

4. Project Scope

4.1 In Scope

- User registration, login, and profile management (User and Admin roles)
- Plant species database integrated via free API with care information
- Personal plant collection management — add, edit, upload photo, archive plants
- Automated care schedule calculation and reminder system
- Care activity logging and plant health journal with notes
- Dashboard with summary of active plants, pending tasks, and care history
- Email and in-app notifications for due tasks
- Admin panel for managing plant database reference and system settings
- Responsive web design compatible with desktop and mobile browsers
- Plant diagnosis via image recognition or AI

4.2 Out of Scope

- Mobile application installation (Android/iOS) — web-based only
- E-commerce or plant purchasing functionality

- Real-time sensor integration or IoT device connectivity
- Social media sharing or community forum features
- Third-party paid licensing costs or proprietary database subscriptions
- Hardware procurement or equipment provision

5. System Features / Functions

Module	Feature	Description
User Management	Registration & Login	Allows users to create personal accounts and securely log in; supports Admin and Regular User roles.
User Management	Profile Management	Users can view and update personal information, set notification preferences, and manage account settings.
Plant Library	Plant Species Database	Displays plant profiles with scientific/common names, care requirements, light needs, and watering frequency — populated via free plant API.
My Plants Collection	Add & Manage Plants	Users add their own plants: upload photo, set location/placement, select species, and assign custom name.
Care Scheduling	Automated Care Reminders	System calculates watering, fertilizing, and repotting schedules; sends email/in-app notifications when tasks are due.
Care Logging	Daily Care Journal	Users log completed tasks (watered, pruned, repotted, etc.); add notes and photos to track plant growth over time.
Plant Health Tracker	Condition Updates & Diagnosis	Users record plant health status; system provides basic tips for

		common issues (yellow leaves, pests, wilting).
Reporting	Dashboard & Progress Reports	Visual summary: active plants, tasks completed, upcoming schedule, growth timeline, and monthly care activity.
Notifications	Care Alerts & Reminders	Sends real-time reminders via email or browser notification for due and overdue plant care tasks.
Administration	System Settings	Allows admin to manage plant species database, moderate user reports, configure notification templates, and manage system parameters.

6. Technology Stack

Layer	Technology / Tool	Purpose
Frontend (Web)	HTML5, CSS3, JavaScript / React.js	Responsive user interface — plant gallery, care calendar, journal forms, dashboard.
Backend / API	Node.js + Express.js	Server-side logic: user authentication, schedule calculation, API data handling, business rules.
Database	MongoDB / MySQL	Stores user accounts, plant profiles, care logs, schedule data, and uploaded images.
API Integration	Trefle API / Perennial API (Free Tier)	Provides plant species details, care guides, and botanical information.
Cloud / Hosting	Render + Netlify / Firebase Hosting	Deployment and hosting environment for web app and backend services.
Version Control	Git + GitHub	Source code management and collaboration.
Project Management Tool	Trello / Notion	Task tracking, progress monitoring, and team collaboration.
Other Tools	Postman (API testing),	API endpoint testing; interface

	Figma (UI design), Cloudinary (image storage)	design; secure storage for plant photos.
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7. Target Users

User Role / Group	Description	Key Needs / Usage
Plant Enthusiasts / Hobbyists	Individuals growing houseplants, herbs, or small gardens at home	Track watering schedules, log plant growth, receive reminders, learn proper care.
Beginner Gardeners	New to plant care, unsure about watering frequency or plant needs	Access plant database, get basic care guides, follow step-by-step reminders.
Home & Urban Gardeners	People with limited space — balconies, windowsills, indoor gardens	Manage multiple plants in different locations, organize care tasks efficiently.
Plant Shop / Nursery Staff	Small business owners managing display plants or customer orders	Track inventory health, maintain care schedules for display stock.
System Administrator	IT student or faculty managing the system	Full access to manage plant database, moderate content, and configure system settings.

8. Project Timeline / Work Plan

Phase	Activities	Estimated Duration
Planning & Requirements Gathering	Survey target users, interview plant enthusiasts, finalize requirements documentation, identify API sources	2 weeks
System Design	UI/UX wireframing, database schema	2 weeks

	design, architecture planning, API integration planning	
Development	Frontend interface development, backend API & authentication, database integration, plant API connection, reminder system	5–6 weeks
Testing & QA	Unit testing, integration testing, user acceptance testing with plant enthusiast participants, bug fixes	2 weeks
Deployment	Domain setup, cloud hosting configuration, data population, user guide preparation, system go-live	1 week
Project Presentation	Final demonstration, defense, and ongoing system maintenance / warranty period	Ongoing / 3 months